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# STM32 USB CDC Example – Virtual COM Port Communication (No UART Needed)

This tutorial explains how to set up a STM32 USB CDC (Virtual COM Port) for bidirectional communication with a PC—**no UART required**. You'll learn how to use the STM32CubeMX tool to configure USB as a Communication Device Class (CDC) and implement both **CDC_Transmit_FS** and **CDC_Receive_FS** to send and receive data directly via USB. This is a perfect method for STM32 **USB virtual COM port applications** like logging, file transfer, and serial communication over USB—especially useful when UART pins are limited or unavailable.

Up to this point, we have relied on UART communication to interface with the computer, enabling the transmission and reception of useful data. Many STM32 development boards, such as the Nucleo and Discovery series, include an on-board ST-Link debugger, which also provides a Virtual COM Port. This feature simplifies the process of sending data to the computer over UART without the need for additional external hardware.

However, when working with custom boards—such as the popular Bluepill board—there is no built-in ST-Link. In such cases, an external USB-to-UART converter module is typically required to enable data transfer between the microcontroller and the computer.

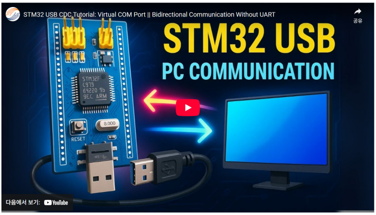
In this tutorial, we will explore how to utilize the USB peripheral of STM32 to communicate directly with a PC, eliminating the need for UART. You will learn how to configure STM32 USB to both send and receive data, in a manner similar to traditional UART communication.

ADVERTISEMENT

### STM32 USB PC Communication Video Tutorial

This tutorial explains how to set up USB CDC on STM32 to communicate directly with your PC, without using UART. To make it easier to follow, I've recorded a step-by-step video walkthrough showing the CubeMX configuration, code changes, and a live demo of sending and receiving data. Watch the video while working with the example code to clearly understand each step of the process.

Watch the USB CDC Tutorial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PC COMMUNICATION
RESET
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## Introduction to STM32 USB Communication

When working with STM32 microcontrollers, we often use **UART** to send and receive data between the board and a PC. However, STM32 also has a powerful feature called **USB Communication Device Class (USB CDC)**. With USB CDC, your STM32 can communicate directly with a computer over USB, just like a virtual COM port, without needing a USB-to-UART converter. This makes STM32 USB communication faster, more reliable, and easier to set up for many IoT and embedded projects.

### Why use USB instead of UART for PC communication

Traditionally, developers use **UART with a USB-to-serial adapter** to connect STM32 to a PC. While this works, it adds extra hardware and limits speed. By using **STM32 USB CDC**, you connect the board directly to your PC's USB port. This approach removes the need for converters, reduces wiring, and offers **higher data transfer rates**. It also allows you to run multiple virtual COM ports if your project requires more than one channel of communication. In short, USB provides **cleaner, faster, and more efficient communication** than UART when working with STM32.

### What is USB CDC (Communication Device Class)

**USB CDC**, or Communication Device Class, is a standard USB protocol that lets devices act as virtual serial ports. When you enable USB CDC on STM32, your computer recognizes the board as a **Virtual COM Port (VCP)**. This means you can use the same serial terminal software you use for UART—like PuTTY, Tera Term, or Arduino Serial Monitor—to exchange data over USB. For developers, this makes the transition from UART to USB seamless, while unlocking the benefits of direct USB communication.

Features of USB CDC in STM32:

1. Virtual COM Port Emulation:
   Creates a virtual serial port on the PC, enabling standard serial communication without physical UART hardware.
2. High-Speed Data Transfer:
   Supports faster data rates compared to traditional UART, limited only by USB specifications.
3. Plug-and-Play Support:
   Works seamlessly with most operating systems (Windows, Linux, macOS) without requiring specialized drivers.
4. Bidirectional Data Exchange:
   Enables simultaneous data transmission and reception, making it suitable for real-time communication applications.

## STM32 USB CDC Project Requirements

Before we begin building our **STM32 ESP8266 MQTT project**, let's take a quick look at the tools and components you'll need. This includes both hardware and software required for the project.

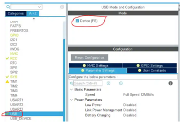
- Hardware:
  - STM32F103 Dev Board
  - ST-Link V2
  - USB Cable
- Software:
  - STM32CubeIDE
  - Serial Monitor

## STM32 USB CDC CubeMX Configuration

Before you can start sending and receiving data with STM32 over USB, you need to configure the USB peripheral in **STM32CubeMX**.

### Enabling USB peripheral in STM32CubeMX

First, open **STM32CubeMX** and select your target STM32 microcontroller. In the **Peripherals** tab, locate and enable the **USB Device (FS)** or **USB OTG FS** peripheral, depending on your board. Make sure to choose **Device Only** mode, since we are using STM32 as a USB device connected to a PC.



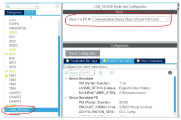
☑ Device (FS)

Configure the below parameters :

Basic Parameters
Speed Full Speed 12MBit/s
Power Parameters
Low Power Disabled
Link Power Management Disabled
Battery Charging Disabled

### Configuring USB as a Virtual COM Port

Next, configure the USB device to work as a **Communication Device Class (CDC)**. In CubeMX, go to the USB configuration settings and select **CDC (Communication Device Class)**. This tells STM32 to act like a **Virtual COM Port (VCP)**, making it appear as a serial port on your PC. You can also adjust endpoint sizes and buffer settings here to optimize data transfer for your application.



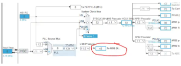
Class For FS IP Communication Device Class (Virtual Port Com)

Configure the below parameters :

Device Descriptor
VID (Vendor IDentifier) 1155
LANGID_STRING (Langua... English(United States)
MANUFACTURER_STRIN... STMicroelectronics
Device Descriptor FS
PID (Product IDentifier) 22336
PRODUCT_STRING (Prod... STM32 Virtual ComPort
CONFIGURATION_STRIN... CDC Config

### Setting the STM32 USB Clock

A correct **USB clock configuration** is essential for reliable STM32 USB communication. The USB peripheral requires a precise 48 MHz clock to function properly. Without this, your **USB CDC virtual COM port** may not work, and data transmission can fail.



## STM32 USB CDC Code Implementation

To send data to a PC using USB CDC, we need to use specific functions provided in the STM32 USB middleware. These functions are implemented in the following file:

```
USB_DEVICE → App → usbd_cdc_if.c
```

### Using `CDC_Transmit_FS` to Send Data to PC

Once your **STM32 USB CDC** is set up, you can start sending data from the microcontroller to your PC using the `CDC_Transmit_FS` function. This function is part of the **STM32 HAL USB CDC library** and makes sending messages over USB straightforward.

How `CDC_Transmit_FS` Works

The `CDC_Transmit_FS` function sends a buffer of data from your STM32 to the PC through the virtual COM port. Its basic syntax looks like this:

```c
uint8_t CDC_Transmit_FS(uint8_t* Buf, uint16_t Len);
```

- `Buf` – Pointer to the data buffer you want to send.
- `Len` – Number of bytes to transmit.

When called, `CDC_Transmit_FS` queues the data for transmission over USB. Your PC will receive this data just like it would from a normal serial port.

### Example Code for Data Transmission

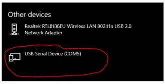
```c
#include "string.h"

/* Private user code ---------------------------------------------------------*/
/* USER CODE BEGIN 0 */
uint8_t *data = "Hello World from USB CDC\n";
/* USER CODE END 0 */

int main(void)
{
  HAL_Init();
  SystemClock_Config();
  MX_GPIO_Init();
  MX_USB_DEVICE_Init();   // Initialize the USB device stack

  while (1)
  {
    CDC_Transmit_FS(data, strlen(data)); // Send data via USB
    HAL_Delay(1000);                      // Wait for 1 second
  }
}
```

### Step-by-Step Explanation of the Code

1. **Include Headers:**
   The required header files ( `usb_device.h` , `usbd_cdc_if.h` , and `string.h` ) are included to use USB functions and string operations.
2. **Define Data Buffer:**
   We define a pointer `data` containing the string `"Hello World from USB CDC\n"` , which will be transmitted.
3. **Initialization:**
   - `HAL_Init()` – Initializes the HAL Library.
   - `SystemClock_Config()` – Configures the system clock.
   - `MX_GPIO_Init()` – Initializes the GPIOs.
   - `MX_USB_DEVICE_Init()` – Sets up the USB device for CDC communication.
4. **Data Transmission Loop:**
   Inside the `while(1)` loop, the function `CDC_Transmit_FS()` sends the content of `data` every 1 second using `HAL_Delay(1000)` .

### Verifying Data on the PC

After you run your STM32 code, the **PC automatically detects the board as a Virtual COM Port (VCP)**. This confirms that the USB CDC setup is working correctly. You can now view the data sent from your STM32 using any serial terminal software.

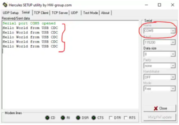


Other devices

Realtek RTL8188EU Wireless LAN 802.11n USB 2.0 Network Adapter

USB Serial Device (COM5)

- When the code runs on the MCU, the PC detects the STM32 as a Virtual COM Port (VCP).
- Open any serial monitor software (e.g., Hercules, Tera Term, or PuTTY) and select the correct COM port.
- You will see the message **"Hello World from USB CDC"** printed every second.



```
Serial port COM5 opened
Hello World from USB CDC
Hello World from USB CDC
Hello World from USB CDC
Hello World from USB CDC
Hello World from USB CDC
```

Name: COM5

Baud: 115200

Data size: 8

Parity: none

Handshake: OFF

Mode: Free

## Using `CDC_Receive_FS` to receive Data from PC

After sending data from STM32 to your PC, you often need to **receive data back from the PC**. The STM32 HAL library provides the `CDC_Receive_FS` function to handle incoming data over USB CDC. This function lets your board act like a true serial device, reading messages sent from the computer in real time.

### How `CDC_Receive_FS` Works

The `CDC_Receive_FS` function is a callback that automatically executes when the PC sends data. CubeMX-generated code links this function to the **USB CDC driver**, so you don't need to manage low-level USB packets manually.

Basic usage example:

```c
uint8_t USB_RxBuffer[64]; // Buffer to store received data

uint8_t CDC_Receive_FS(uint8_t* Buf, uint32_t *Len) {
    memcpy(USB_RxBuffer, Buf, *Len); // Copy received data to buffer
    // Process the received data here
    return USBD_OK;
}
```

- `Buf` – Points to the data received from the PC.
- `Len` – Number of bytes received.
- Return `USBD_OK` to signal successful reception.

### Example: Echoing Data Back to PC

You can combine `CDC_Receive_FS` with `CDC_Transmit_FS` to **echo messages back to the PC**, which is useful for testing:

```
uint8_t CDC_Receive_FS(uint8_t* Buf, uint32_t *Len) {
    // Copy received data to buffer
    memcpy(USB_RxBuffer, Buf, *Len);
    // Send it back to PC
    CDC_Transmit_FS(Buf, *Len);
    return USBD_OK;
}
```

This setup allows STM32 to **receive any message typed on the PC** and immediately send it back, confirming two-way USB communication.

### Tips for Using `CDC_Receive_FS`

1. **Use a sufficient buffer** – Ensure the buffer can hold the largest expected message from the PC.
2. **Process data quickly** – Avoid long delays inside the callback to prevent USB communication issues.
3. **Combine with interrupts** – For continuous communication, process received data in the main loop or in a separate task after copying it from the callback.

Using `CDC_Receive_FS` makes your **STM32 USB CDC communication fully bidirectional**, eliminating the need for UART while enabling real-time interaction with your PC.

## Advantages of STM32 USB CDC over UART

Using **STM32 USB CDC** instead of traditional UART offers several important benefits. It makes your PC communication faster, cleaner, and more flexible.

### Faster Data Transfer Speed

USB CDC provides **higher data transfer rates** compared to UART. With USB, STM32 can send and receive larger amounts of data quickly, making it ideal for applications that require frequent or high-volume communication. This speed improvement helps reduce latency and ensures smoother data logging and control tasks.

### Direct PC Connectivity (No USB-UART Converter Needed)

Unlike UART, USB CDC allows the STM32 to **connect directly to a PC**. You don't need a USB-to-UART converter or extra wiring. The STM32 appears as a **Virtual COM Port** on the PC, so you can use standard serial terminal software like PuTTY, Tera Term, or Hercules. This simplifies setup and reduces hardware costs.

### Reliable Bidirectional Communication

USB CDC supports **full bidirectional communication**, meaning you can send data from STM32 to the PC and receive data back at the same time. Unlike UART, which may require additional circuitry for multiple channels, USB CDC provides a **stable and reliable two-way communication channel** out of the box.

## Common Issues and Troubleshooting STM32 USB CDC

Even with a correct setup, you may face some common issues when using USB CDC on STM32. Here's how to identify and fix them.

### Virtual COM Port Not Showing Up on PC

If your PC does not detect the STM32 as a Virtual COM Port:

- Check that the **USB cable is working** and supports data transfer (some cables are power-only).
- Ensure the **USB peripheral is enabled** in CubeMX.
- Verify that the **USB drivers are installed** on your PC.

### Data Not Transmitting or Receiving Correctly

If data does not appear on the PC or seems corrupted:

- Confirm that you are using the **correct COM port and settings** in your terminal software.
- Make sure the STM32 code calls `CDC_Transmit_FS` and `CDC_Receive_FS` correctly.
- Avoid sending data too quickly without checking for **previous transmission completion**.

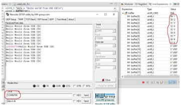
### Driver and Firmware Related Issues

Some problems arise due to outdated or missing drivers:

- Update your **USB VCP drivers** for Windows, macOS, or Linux.
- Ensure CubeMX generates the **latest USB CDC firmware** for your STM32.
- If using custom USB code, verify that **endpoint sizes, buffer lengths, and descriptors** are configured correctly.

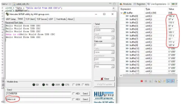
## STM32 USB CDC Result

The result of the above code is shown in the images below



```
Hello World from USB CDC
Hello World from USB CDC
Hello World from USB CDC
Hello World from USB CDC
Hello World from USB CDC
Hello World from USB CDC
Hello World from USB CDC
123456789Hello World from USB CDC
Hello World from USB CDC
Hello World from USB CDC
Hello World from USB CDC
Hello World from USB CDC
Hello World from USB CDC
```

The controller was keep transmitting "**Hello World from USB CDC**". When the PC sent the data "123456789" (Represented by Pink colour on the console) to the controller, it gets stored in the **buffer**.
You can see the data stored in the buffer in the debugger's Live Expression window as shown in the image above.



```
uint8_t *data = "Hello World from USB CDC\n";
uint8_t buffer[64];
```

```
Hello World from USB CDC
Hello World from USB CDC
Hello World from USB CDC
data is okHello World from USB CDC
Hello World from USB CDC
```

The same behavior was observed in this case as well. This time, a different set of data was transmitted, and it was successfully stored in the buffer.

Since the buffer is accessible from the `main.c` file, the received data can be processed further in any desired manner, such as parsing commands, logging information, or triggering specific application tasks.

## Conclusion

Using STM32 USB CDC allows you to communicate with a PC without using UART, making your projects faster, cleaner, and more reliable. By setting up the USB peripheral in STM32CubeMX, configuring it as a Virtual COM Port, and using `CDC_Transmit_FS` and `CDC_Receive_FS`, you can send and receive data seamlessly.

USB CDC offers higher data transfer speeds, direct PC connectivity, and stable bidirectional communication, making it a superior choice over traditional UART connections.

Even if you encounter issues like the Virtual COM Port not showing or data transmission problems, careful configuration and troubleshooting ensure smooth USB communication.

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## STM32 USB CDC FAQs

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* * *

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Ali

10 months ago

Hello Sir, hope you are doing well, i followed the steps from tutorial but the device is not showing up in com port of PC, i have tried different codes like led blink and all which works fine but the CDC doesn't work , the code gets uploaded successfully , but no com port show on device manager . pls help.

```
Log output file: C:\Users\NW\AppData\Local\Temp\STM32CubeProgrammer_a05264.log
ST-LINK SN : 211405003212364D434B4E00
ST-LINK FW : V2J45S7
Board : —
Voltage : 3.19V
SWD freq : 4000 KHz
Connect mode: Under Reset
Reset mode : Hardware reset
Device ID : 0x410
Revision ID : Rev X
Device name : STM32F101/F102/F103 Medium-density
Flash size : 64 KBytes
Device type : MCU
Device CPU : Cortex-M3
BL Version : —

Opening and parsing file: ST-LINK_GDB_server_a05264.srec

Memory Programming …
File : ST-LINK_GDB_server_a05264.srec
Size : 29.15 KB
Address : 0x08000000

Erasing memory corresponding to segment 0:
Erasing internal memory sectors [0 29]
Download in Progress:

File download complete
Time elapsed during download operation: 00:00:01.887

Verifying …

Download verified successfully

Shutting down…
Exit.
```

+ 0 − Reply

saeed

1 year ago

Hi,
Thanks for the tenurial you have made.
please make and share another one for **USB BULK TRANSFER** application if it is viable for you.
many thanks

+ 0 − Reply

Mohammad

1 year ago

Thank you so much 🙏

+ 0 − Reply

mjh

1 year ago

I'm using a blue pill with stm32f103c8t6 using STlink-V2 when I flash the file to the blue pill the stlink disconnected and not able to debug and see the data I sent to the board.
How did you connect your blue pill board and stlink.
Very good tutorial.

+ 0 − Reply

Mehrdad

1 year ago

Hi, how should I receive unknown length of data from usb com port?I know that such operation in uart could be done using Idle line detector. but how about usb cdc receive?

+ 0 − Reply

ControllersTech Author

*Reply to Mehrdad* 1 year ago

The function int8_t CDC_Receive_FS(uint8_t* Buf, uint32_t *Len) already receives the data of unknown length. Here Len is the pointer to the number of data byte received.

+ 0 − Reply

Andy

1 year ago

after I press the reset button, data is no longer received by Hercules. I had to reconnect the USB to get it working again. Is there a way to prevent that?

+ 0 − Reply

neeraj

2 years ago

i am using STM32F413zh nucleo when i connect the micro-USB cable to PC there is nothing as appearance in device manager????
i connect D+ and D- in p12 and p11.
in device manage only st-link cable was showing.
what i do please help….

+ 0 − Reply

ControllersTech Author

*Reply to neeraj* 2 years ago

1. Why are you even doing this on the nucleo board ? You can simply send the data to the PC via the STlink using the UART2.
2. Nucleo STM32F413zh has one user USB at the bottom. Use it.

+ 0 − Reply

Karim

3 years ago

Salut, j'utilise un STM32H735G-DK. J'ai suivi votre tuto, et vu que les désignations de fonction diffèrent (STM32F & STM32H) j'ai du modifier le "F" de CDC_Transmit_FS par "H" ce qui m'a donné CDC_Transmit_HS. Malgré toutes mes tentatives, mon pc ne détecte pas le port vituel de mon MCU. Svp j'ai besoin d'aide . . . vous pouvez me contacter à mon adresse m.a.i.l: karimnana520 @gmail.com ou sur mon whatsapp actuel: +237 658747103

+ 0 − Reply

ControllersTech Author

*Reply to Karim* 3 years ago

USB HS Does not supports virtual Communication. You have to use FS only

+ 0 − Reply

babak

3 years ago

I'm using stm32f429zit6 but my board is not detected by hercule. what am i do?

+ 0 − Reply

**ControllersTech** Author

*Reply to babak* 3 years ago

if you are using the nucleo or discovery boards, make sure you connect to the USER USB, not the ST-Link
Also check if the cable supports the data transfer or not

+ 0 − Reply

**RyanSIlvers**

3 years ago

This tutorial in Stdlibrary instead please

+ 0 − Reply

**12val12**

4 years ago

How recieve packet from PC

B0=254; // sync
B1 //number of variable
B2//Low byte
B3// byte
B4// byte
B5=//High Byte
Var[B1]=B2+B3<<8+B4<<16+B5<<32
If the stream fails, then in the incoming stream we look for synchrobyte 254 (skip bytes until we have received 254)

+ 0 − Reply

**Nguyễn Tuấn Anh**

4 years ago

how can i transmit a HEX value instead of char?
Thanks!

+ 0 − Reply

**ControllersTech** Author

*Reply to Nguyễn Tuấn Anh* 4 years ago

just send it simply. What's the issue?

+ 0 − Reply

**rajkumar**

4 years ago

Hai, how can we send/transmit the adc value through the usb cdc virtual comfort, I am new to stm32 ,if anyone known please help me

+ 0 − Reply

**ControllersTech** Author

*Reply to rajkumar* 4 years ago

try sending the address to the value.
If can't, then convert it to string and send

+ 0 − Reply

**Costa**

*Reply to rajkumar* 3 years ago

https://www.digikey.com/en/maker/projects/getting-started-with-stm32-working-with-adc-and-dma/f5009db3a3ed4370acaf545a3370c30c

+ 0 − Reply

**ahmed**

4 years ago

stm32f4 ws2812b with cubeide.

+ 0 − Reply

**Tan Sang**

4 years ago

That great! You can make video with NRF24L01, please!

+ 0 − Reply

**reza**

4 years ago

it's great 😍

*Last edited 4 years ago by reza*

+ 4 − Reply

**Peter**

*Reply to reza* 4 years ago

The USB_CDC_ReceivePacket is the acknowledge of receiving. It should be called after copy the data.

+ 0 − Reply

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